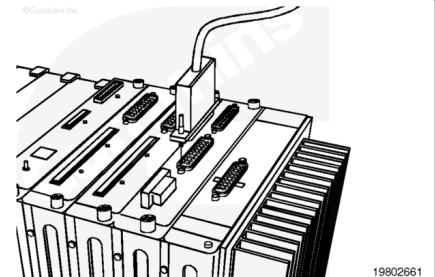


Trainings ▾

Pin Replacement

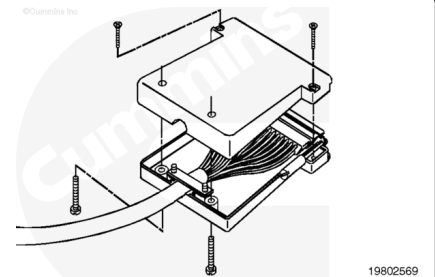
The D-sub miniature connector is used to attach the appropriate harnesses to the ECM.

Remove the connector.

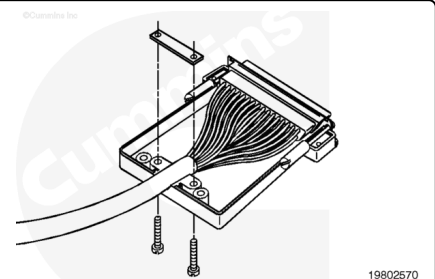


Remove the backshell from the connector.

Locate the damaged pin or wire.



Remove the strain relief to be able to work with the damaged pin.



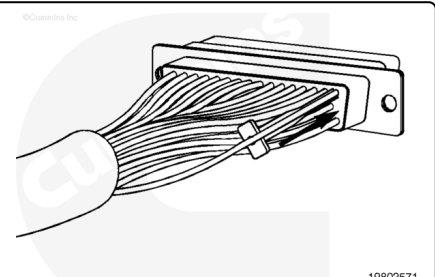
Use the D-sub miniature extraction tool, Part Number 3163971, place over the wire to remove a pin from the connector.

Before installing the new repair wire, perform a test fit to make sure the wire is the correct size.

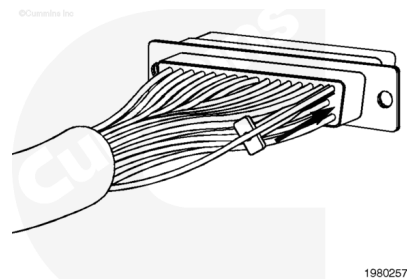
Refer to the appropriate wiring repair kit in the service tools table in the front of Section 19 for the correct repair wire.

Replace one contact wire at a time. If more than one wire needs replaced, attach a lettered tag to each wire removed.

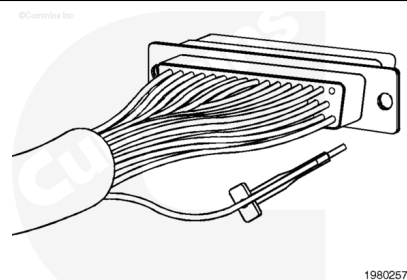
Refer to the wiring diagram in Section E for pin locations.



With a twisting motion, push the pin extraction tool into the connector approximately 25 mm [1 in] until it bottoms on the terminal flange. A click will be heard when the extraction tool is in place.



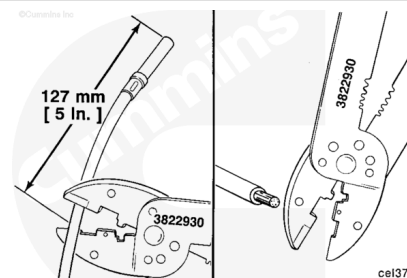
Hold the tool on the terminal flange and pull the wire and connecting pin out of the connector. Note and record the hole from which the pin is removed.



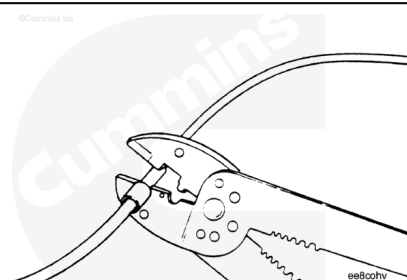
Note : The repair wire is 127 mm [5 in] long.

Use wire crimping tool, Part Number 3822930, to cut 127 mm [5 in] off the wire and pin.

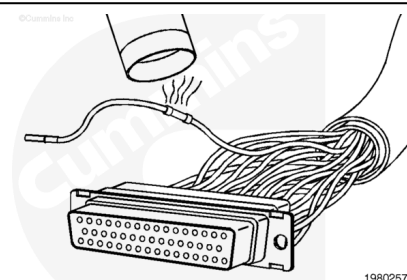
Use the crimping tool to remove 6 mm [$\frac{1}{4}$ in] of insulation from the wire.



Use the wire crimping tool to crimp the repair wire onto the bare wire.



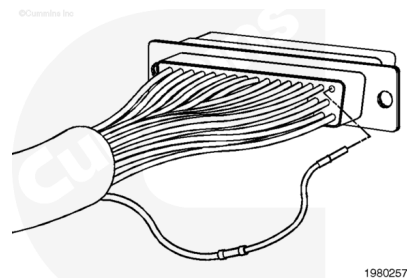
Use heat gun, Part Number 3822860, to heat shrink the tubing around the wire. The tubing will shrink and make the connection waterproof.



Insert the pin into the correct hole of the connector.

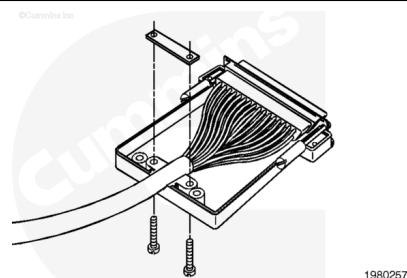
The pin **must** click into place and hold the wire in the connector.

Pull the wire gently to make sure it is seated in the connector.



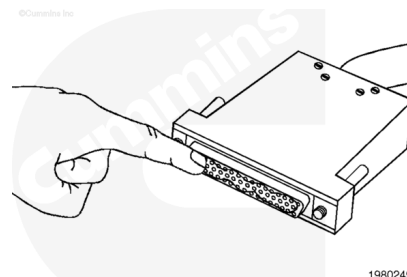
Tighten the strain relief.

Complete the assembly of the backshell by placing the two halves together and tightening the screws.



⚠ CAUTION ⚠

Use only Cummins-recommended lubricant DS-ES, Part Number 3822934. Other lubricants, such as lubricating oil or grease, in the connectors can cause ECM damage, poor engine performance, or premature connector pin wear.



Apply a small amount of lubricant to the connector terminals.
Do **not** fill the entire connector cavity with lubricant.

Connector Replacement

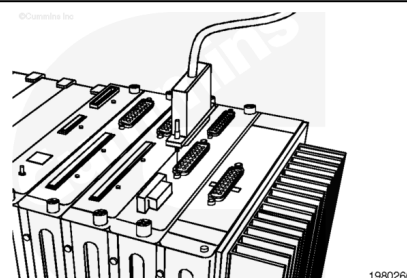
The 50-pin D-sub miniature connector is used to attach the appropriate harnesses to the ECM.

Before installing the new connector, perform a test fit to make sure the connector is keyed correctly.

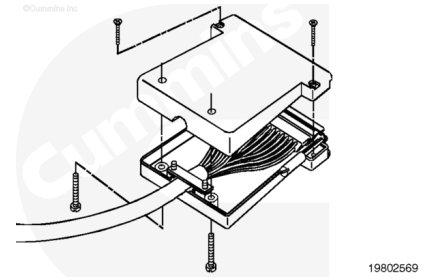
Refer to the appropriate wiring repair kit in the service tools table in the front of Section 19 for the correct repair connector.

Refer to the wiring diagram in Section E for pin locations.

Replace one contact wire at a time. If more than one wire needs replaced, attach a lettered tag to each wire removed.



Remove the backshell from the connector. Remove the strain relief.



To replace the connector use the D-sub miniature extraction tool, Part Number 3163971, place over each wire to remove all pins from the connector.

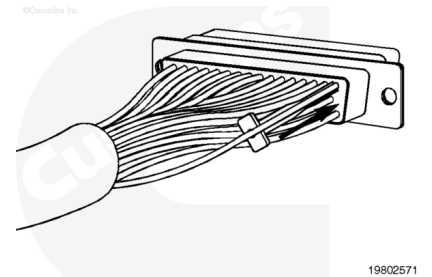
Before installing the new connector, perform a fit test to make sure the connector is keyed correctly.

Refer to the appropriate wiring repair kit in the service tools table in the front of Section 19 for the correct repair connector.

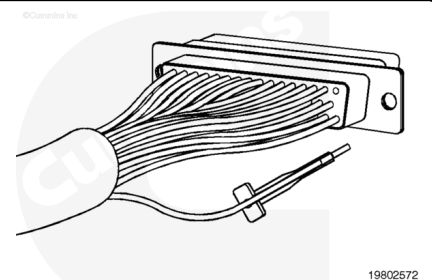
Refer to the wiring diagram in Section E for pin locations.

Replace one contact wire at a time. If more than one wire needs replaced, attach a lettered tag to each wire removed.

With a twisting motion, push the pin extraction tool into the connector approximately 25 mm [1 in] until it bottoms on the terminal flange. A click will be heard when the extraction tool is in place.



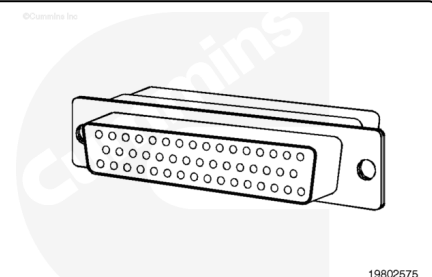
Hold the tool on the terminal flange and pull the wire and connecting pin out of the connector. Note and record the hole from which the pin was removed.



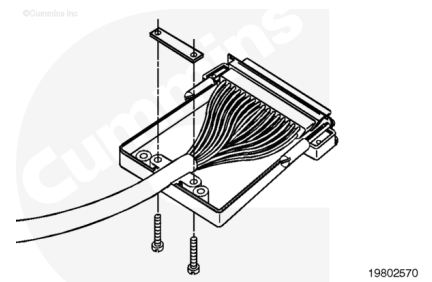
Insert the pins into the correct holes of the replacement connector.

Each pin **must** click into place and hold the wires in the connector.

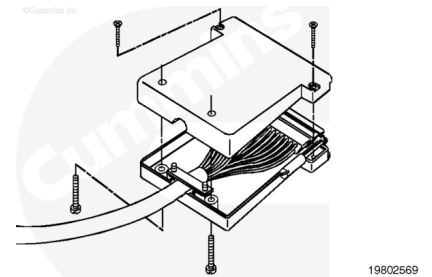
Pull each wire gently to make sure it is seated in the connector.



Place the connector pin block onto the lower half of the backshell. Place all wires within the strain relief and tighten the strain relief.

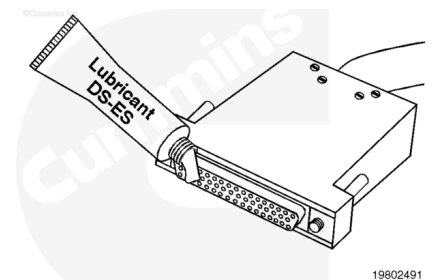


Complete the assembly of the connector by placing the upper half of the backshell onto the lower half and tightening the screws.



⚠ CAUTION ⚠

Use only Cummins-recommended lubricant DS-ES, Part Number 3822934. Other lubricants, such as lubricating oil or grease, in the connectors can cause ECM damage, poor engine performance, or premature connector pin wear.



Apply a small amount of lubricant to the connector terminals. Do **not** fill the entire connector cavity with lubricant.